KALI LINUX FIELD REFERENCE GUIDE

OSI, TCP/IP, DNS, HTTP/S & Subnetting

Comprehensive Architecture, Protocol Analysis & Network Calculations

// # 1. OSI Model Layers & Functions

The Open Systems Interconnection (OSI) model standardizes network communication into 7 distinct abstract layers.

L#	Layer Name	Core Function	PDU	Key Protocols
7	Application	Network services directly to end-user applications & processes.	Data	HTTP/S, DNS, SSH, FTP, SMTP
6	Presentation	Data translation, encryption/decryption, compression, formatting.	Data	SSL/TLS, JPEG, ASCII, SSH
5	Session	Establishes, manages, and terminates session connections.	Data	NetBIOS, PPTP, RPC, SOCKS
4	Transport	End-to-end communication, error recovery, flow control, segmentation.	Segment / Datagram	TCP, UDP
3	Network	Logical addressing (IP) and packet routing across networks.	Packet	IPv4, IPv6, ICMP, IPsec, OSPF
2	Data Link	Physical addressing (MAC), node-to-node transfer, error detection.	Frame	Ethernet, Wi-Fi (802.11), ARP
1	Physical	Transmission and reception of raw bitstreams over physical medium.	Bit	Fiber, Copper, Hubs, Repeaters

// # 2. TCP/IP Protocol Suite & Transport Layer

TCP Transmission Control Protocol

- **Connection-Oriented:** Requires explicit session establishment.
- **Reliable:** Retransmits lost packets & guarantees in-order delivery.
- **3-Way Handshake:**

```
Client  -- SYN -->      Server
Client  <-- SYN-ACK --  Server
Client  -- ACK -->      Server
```

- **Header Size:** 20–60 Bytes.
- **Use Cases:** Web (HTTP/S), SSH, FTP, Email (SMTP).

UDP User Datagram Protocol

- **Connectionless:** No handshake or state tracking.
- **Unreliable:** Best-effort delivery; no retransmission or ordering.
- **Low Overhead / Speed:** Minimal header size (8 Bytes) for high throughput.
- **Stateless Structure:**

```
Client  -- Request Datagram --> Server
Client  <-- Response Datagram -- Server
```

- **Use Cases:** DNS, VoIP, Video Streaming, DHCP, NTP.

// # 3. DNS & HTTP/HTTPS Deep Dive

Domain Name System (DNS) Resolution Chain

DNS translates human-readable domain names into IP addresses via a recursive lookup flow:

1. **Client Query:** Local host checks OS DNS cache / `/etc/hosts`. If missing, sends request to Recursive Resolver (e.g., `1.1.1.1`).
2. **Root Server (`.`):** Directs the resolver to the appropriate Top-Level Domain (TLD) server.
3. **TLD Server (`.com`):** Directs resolver to domain's Authoritative Name Server.
4. **Authoritative Name Server:** Returns requested target record (A for IPv4, $AAAA$ for IPv6).

HTTP vs. HTTPS (TLS/SSL Security)

HTTP (Port 80): Plaintext application communication. Susceptible to interception and spoofing.

HTTPS (Port 443): HTTP wrapped inside a TLS (Transport Layer Security) encrypted tunnel.

> **TLS 1.3 Handshake Protocol:**
> 1. `ClientHello`: Supported cipher suites & key exchange parameters.
> 2. `ServerHello`: Selected cipher suite, Server Certificate, and Key Share.
> 3. `Key Calculation`: Both parties derive symmetric keys (1-RTT completed).
> 4. `Encrypted Traffic`: Application data payload is transmitted securely.

// # 4. IP Addressing, Subnetting, and NAT

RFC 1918 Private IPv4 Ranges

- **Class A:** `10.0.0.0/8` (10.0.0.0 – 10.255.255.255) → 16.7M hosts
- **Class B:** `172.16.0.0/12` (172.16.0.0 – 172.31.255.255) → 1M hosts
- **Class C:** `192.168.0.0/16` (192.168.0.0 – 192.168.255.255) → 65k hosts

Subnetting Calculations

For a given CIDR network prefix length n:

$$Total\ Usable\ Hosts = 2^{(32 - n)} - 2$$

(Subtract 2 for Network Address and Broadcast Address)

CIDR Prefix	Subnet Mask	Total Addresses	Usable Host IPs
`/24`	`255.255.255.0`	256	254
`/27`	`255.255.255.224`	32	30
`/28`	`255.255.255.240`	16	14
`/30`	`255.255.255.252`	4	2 (Point-to-Point)

Network Address Translation (NAT) Modes

- **Static NAT:** Maps a single private IP address to a dedicated public IP address (1-to-1).
- **Dynamic NAT:** Maps private IPs to an available pool of public IP addresses.
- **PAT (Port Address Translation / NAT Overload):** Maps multiple private IP addresses to a single public IP using unique source port numbers.

// # 5. Kali Linux Command Cheatsheet

Terminal - Network Inspection Commands

```
# View interface addresses and link status
ip -c a

# Trace routing path & hop latency
traceroute example.com

# Inspect active TCP/UDP connections & listening ports
ss -tulpn

# DNS record lookup via recursive query
dig A example.com +trace

# Live packet capture on eth0 interface
tcpdump -i eth0 -nn -s 0 -v "tcp port 80 or tcp port 443"
```